Structured Denoising with EM-BP and One-Dimensional Gradient SPDEs

A unified guide to Markov score estimation, transfer matrices, exact inference, non-Gaussian structure, and the stochastic Allen–Cahn benchmark

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This document is designed as a self-contained study guide and research blueprint. It distinguishes exact statements, approximations, implementation choices, and open research questions.

Preface

This document consolidates two connected discussions. The first concerned exact score estimation for one-dimensional Markov data using belief propagation (BP), expectation–maximisation (EM), Gaussian closure, neural denoisers, and spectral representations. The second proposed stationary snapshots of one-dimensional gradient stochastic partial differential equations (SPDEs), especially stochastic Allen–Cahn / ϕ^4, as a cleaner and more physically compelling benchmark.

The central scientific programme is:

The programme in one line

Learn a flexible *local probabilistic law*, compose it globally through exact one-dimensional inference, and compare the resulting Bayes-optimal denoiser against Gaussian and neural approximations.

The internal meeting notes record a discussion between Jerome Garnier-Brun and Giovanni Mantovani on EM-BP versus neural denoisers, including the recommendation to lead with structure rather than Fisher information, improve grid BP, establish the Gaussian sanity check, and measure actual inference cost. The notes also warn that they were reconstructed from an automatic transcript, so a few numerical details may be approximate.

How to read this guide

- Part I develops the Markov-chain denoising problem from first principles.
- Part II explains the one-dimensional gradient-SPDE benchmark and why it is especially clean.
- Part III unifies the two stories into a defensible research narrative and experimental plan.
- The appendices collect derivations, pseudocode, notation, and the practical action list.

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1 The thesis in one page

Suppose a clean sequence

$$X = (X_1, \dots, X_L)$$

is first-order Markov:

$$p_\theta(x_1, \dots, x_L) = \mu_\theta(x_1) \prod_{i=1}^{L-1} K_\theta(x_{i+1} \mid x_i).$$

We do not observe X directly. Instead, we observe a Gaussian-corrupted sequence

$$Y_i = X_i + \sigma Z_i, \qquad Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

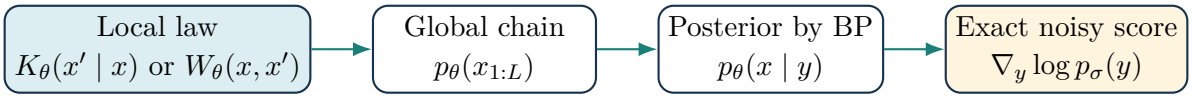
The objective is the score of the noisy distribution,

$$s_\sigma(y) = \nabla_y \log p_\sigma(y).$$

For Gaussian corruption, Tweedie's identity gives

$$s_{\sigma,i}(y) = \frac{\mathbb{E}[X_i \mid Y = y] - y_i}{\sigma^2}.$$

Therefore score estimation reduces to posterior inference. Since the latent graph is a chain, BP computes the posterior means exactly when K_θ is known. If the transition law is unknown, EM can learn it from clean or noisy sequences, depending on which variables are latent.



The Gaussian case is a complete sanity check: BP messages remain Gaussian, the forward pass is a Kalman filter, the full forward–backward pass is a Kalman smoother, and the score is linear. The non-Gaussian case is where structure matters: BP retains multimodality, heavy tails, state dependence, and higher-order information that Gaussian closure removes.

The one-dimensional SPDE proposal strengthens the benchmark. For a reversible gradient SPDE,

$$\partial_t \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\, \eta,$$

the stationary spatial law is

$$\mu(\,\mathrm{d}\phi) \propto \exp\left[-\frac{1}{\Theta} \int \left(\frac{1}{2}(\partial_x \phi)^2 + V(\phi)\right) \mathrm{d}x\right] \mathrm{d}\phi.$$

After discretising space, this is exactly a nearest-neighbour Gibbs measure. BP becomes the transfer-matrix method, exact equilibrium samples can be generated without temporal equilibration, and the oracle score is available for every noise level. A double-well V produces domains and kinks, giving a controlled setting in which Gaussian closure should fail and finite-window neural denoisers should become limited by their receptive fields.

Core contribution

The distinctive message is not merely that higher-order information is erased by noising. It is that one-dimensional probabilistic structure converts a high-dimensional score-estimation problem into learnable local modelling plus exact global inference.

2 Three objects that must not be confused

Much of the conceptual difficulty disappears once three different objects are kept separate.

2.1 The clean process

The clean law is

$$p_\theta(x_{1:L}) = \mu_\theta(x_1) \prod_{i=1}^{L-1} K_\theta(x_{i+1} \mid x_i).$$

The index i can represent time, space, or frame number. In the SPDE benchmark it represents spatial position. The parameter θ controls the transition law.

A simple synthetic example is

$$X_{i+1} = \rho X_i + \varepsilon_i,$$

where the innovations ε_i are chosen to match a Gaussian in mean and variance but differ in fourth-order structure. This isolates information that is invisible to a covariance-only approximation.

2.2 The noisy observation

For additive Gaussian corruption,

$$Y_i = X_i + \sigma Z_i.$$

The likelihood factorises:

$$p_\sigma(y \mid x) = \prod_{i=1}^{L} g_\sigma(y_i \mid x_i), \qquad g_\sigma(y_i \mid x_i) \propto \exp\left[-\frac{(y_i - x_i)^2}{2\sigma^2}\right].$$

Consequently,

$$p_\theta(x \mid y) \propto \mu_\theta(x_1) \prod_{i=1}^{L-1} K_\theta(x_{i+1} \mid x_i) \prod_{i=1}^{L} g_\sigma(y_i \mid x_i)$$

remains a chain.

2.3 The noisy-data score

The denoising score is

$$s_\sigma(y) = \nabla_y \log p_\sigma(y).$$

This is a derivative with respect to the observed data. It must not be confused with the parameter score

$$u_\theta(y) = \frac{\partial}{\partial \theta} \log p_\theta(y),$$

which is used in Fisher information:

$$I(\theta) = \mathbb{E}_\theta[u_\theta(Y)^2].$$

The first object drives denoising and reverse diffusion. The second measures statistical information about a parameter.

Two meanings of "score"

Always state the differentiation variable. In this project, "score estimation" normally means $\nabla_y \log p_\sigma(y)$, while Fisher information uses $\partial_\theta \log p_\theta(y)$.

3 Belief propagation as exact posterior inference

3.1 Forward messages

Define

$$\alpha_i(x_i) = p(y_{1:i}, x_i)$$

up to a normalising constant. Then

$$\alpha_1(x_1) = \mu_\theta(x_1) g_\sigma(y_1 \mid x_1),$$

and

$$\alpha_{i+1}(x_{i+1}) = g_\sigma(y_{i+1} \mid x_{i+1}) \int \alpha_i(x_i) K_\theta(x_{i+1} \mid x_i)\, \mathrm{d}x_i.$$

The update first predicts through the transition and then incorporates the new local observation.

3.2 Backward messages

Define

$$\beta_i(x_i) = p(y_{i+1:L} \mid x_i).$$

Then

$$\beta_L(x_L) = 1,$$

and

$$\beta_i(x_i) = \int K_\theta(x_{i+1} \mid x_i) g_\sigma(y_{i+1} \mid x_{i+1}) \beta_{i+1}(x_{i+1})\, \mathrm{d}x_{i+1}.$$

The backward message summarises all future evidence.

3.3 One-site and pair marginals

The smoothed one-site posterior is

$$p(x_i \mid y) \propto \alpha_i(x_i)\beta_i(x_i).$$

Hence

$$\mathbb{E}[X_i \mid y] = \frac{\int x_i \alpha_i(x_i)\beta_i(x_i)\, \mathrm{d}x_i}{\int \alpha_i(x_i)\beta_i(x_i)\, \mathrm{d}x_i}.$$

The adjacent pair posterior is

$$p(x_i, x_{i+1} \mid y) \propto \alpha_i(x_i) K_\theta(x_{i+1} \mid x_i) g_\sigma(y_{i+1} \mid x_{i+1}) \beta_{i+1}(x_{i+1}).$$

These pair marginals are sufficient for learning any pairwise transition or energy model.

3.4 From posterior mean to exact score

For additive Gaussian corruption,

$$s_{\sigma,i}(y) = \frac{\mathbb{E}[X_i \mid y] - y_i}{\sigma^2}.$$

Thus the exact score follows from a single forward–backward sweep.

Why BP is exact here

BP is exact because the latent factor graph is a tree, specifically a chain. Any remaining error comes from numerical quadrature, state-space truncation, finite precision, or a misspecified transition model, not from loopy-message approximation.

4 Learning the local law with EM

4.1 When EM is actually needed

If clean transitions (X_i, X_{i+1}) are directly observed and the kernel family is straightforward, ordinary maximum likelihood may suffice. EM becomes useful when:

- the clean states are latent because only noisy sequences are observed;
- mixture-component assignments inside K_θ are latent;
- the model contains additional hidden variables;
- the pair law is represented by an energy model whose sufficient statistics must be averaged under a posterior.

4.2 E-step

At the current parameter value θ^{old}, compute

$$\xi_i(x, x') = p_{\theta^{\text{old}}}(X_i = x, X_{i+1} = x' \mid Y = y)$$

using BP. These are soft transition counts.

4.3 M-step for a normalised transition kernel

Update θ by maximising

$$Q(\theta \mid \theta^{\text{old}}) = \sum_i \iint \xi_i(x, x') \log K_\theta(x' \mid x) \, \mathrm{d}x \, \mathrm{d}x'.$$

For multiple sequences, sum this objective over samples.

4.4 Mixture transition model

A flexible conditional mixture may be written as

$$K_\theta(x' \mid x) = \sum_{c=1}^{C} \pi_c(x; \theta) \mathcal{N}\big(x'; \mu_c(x; \theta), s_c^2(x; \theta)\big).$$

The current implementation described in the meeting used grid messages on 401 points over $[-8, 8]$ and a mixture of C Gaussian components. The case $C = 1$ reproduces the Gaussian model. Increasing C increases flexibility but can create narrow spikes, overfitting, and under-resolved grid structure.

Bias–variance–resolution trade-off

A larger mixture is not automatically better. If a component becomes narrower than the grid spacing, the inferred likelihood may improve numerically while the represented transition becomes physically meaningless or unstable.

5 The Gaussian case: score, BP, and Kalman smoothing

5.1 Closed-form score

Assume

$$X \sim \mathcal{N}(0, \Sigma_X), \qquad Y = X + \sigma Z, \qquad Z \sim \mathcal{N}(0, I).$$

Then

$$Y \sim \mathcal{N}(0, \Sigma_X + \sigma^2 I),$$

so

$$s_\sigma(y) = -(\Sigma_X + \sigma^2 I)^{-1} y.$$

The Gaussian posterior mean is

$$\mathbb{E}[X \mid Y = y] = \Sigma_X (\Sigma_X + \sigma^2 I)^{-1} y.$$

Therefore

$$\frac{\mathbb{E}[X \mid y] - y}{\sigma^2} = -(\Sigma_X + \sigma^2 I)^{-1} y,$$

which confirms Tweedie's identity directly.

5.2 Gaussian BP is Kalman inference

For a linear Gaussian transition,

$$X_{i+1} = \rho X_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, q),$$

with Gaussian observations, every message is Gaussian. Each message is determined by a mean and a variance, or equivalently by a precision and an information vector.

The forward BP pass is the Kalman filter. The backward pass combines future evidence with the forward filter. The complete forward–backward computation is therefore a Kalman smoother, more specifically an instance of Rauch–Tung–Striebel smoothing.

Role of the Gaussian experiment

Gaussian grid BP is not computationally attractive; it carries hundreds of values where two parameters would suffice. Its value is diagnostic: the grid implementation should converge to the analytic linear score and the Kalman smoother as the model approaches the Gaussian limit.

5.3 What Gaussian closure means

For a non-Gaussian process, an exact message may be skewed, heavy-tailed, or multimodal. Gaussian closure replaces it by

$$m_i(x) \approx \mathcal{N}(x; \mu_i, v_i),$$

typically by matching first and second moments after each update.

This discards all information beyond mean and covariance. The exact relationship to a "Gaussian score" depends on the projection rule used, so the implementation should state whether the closure is moment matching, assumed-density filtering, expectation propagation, or another approximation.

6 Why non-Gaussian structure matters

6.1 A controlled second-versus-fourth-order experiment

Consider two innovation laws with

$$\mathbb{E}[\varepsilon] = 0, \qquad \mathrm{Var}(\varepsilon) = 1,$$

but different fourth cumulants

$$\kappa_4 = \mathbb{E}[\varepsilon^4] - 3\mathbb{E}[\varepsilon^2]^2.$$

For a Gaussian, $\kappa_4 = 0$. A variance-matched Laplace law has positive excess kurtosis.

If the autoregressive coefficient ρ is unchanged, the resulting chains can have nearly identical covariance structure while differing in tails and local shape. This creates a principled comparison:

- a Gaussian score exploits all second-order structure;
- exact non-Gaussian BP exploits the complete transition law;
- a flexible neural denoiser may learn either, but needs sufficient data and suitable inductive bias.

The meeting notes report approximate relative-score errors of about 0.67 when sequence structure is ignored and 0.168 for the Gaussian score. Those values should be verified against stored outputs because they came from an automatic transcript.

6.2 Hermite modes under Gaussian noising

Under the Ornstein–Uhlenbeck noising semigroup,

$$Y_t = e^{-t} X + \sqrt{1 - e^{-2t}}\, Z,$$

Hermite polynomials are eigenfunctions:

$$P_t H_n = e^{-nt} H_n.$$

Thus degree-n deviations decay as e^{-nt}. In broad terms:

Statistical feature	Typical Hermite degree	Amplitude under noising
Mean	1	e^{-t}
Covariance	2	e^{-2t}
Skewness	3	e^{-3t}
Kurtosis / fourth cumulant	4	e^{-4t}

This explains why fourth-order distinctions become less visible earlier than covariance structure.

6.3 Fourier interpretation

For additive Gaussian noise,

$$p_\sigma = p_0 * \mathcal{N}(0, \sigma^2 I),$$

so

$$\widehat{p}_\sigma(k) = \widehat{p}_0(k) e^{-\sigma^2 \|k\|^2/2}.$$

Large frequencies are exponentially suppressed. Sharp distributional features, narrow peaks, and rapid variations are smoothed first.

Cumulant subtlety

Under pure additive Gaussian noise, unnormalised cumulants of order three and above are not changed by the added Gaussian itself. What decreases is their standardised or observable effect because the total variance increases and the density is smoothed. Under a variance-preserving diffusion, higher cumulants are additionally multiplied by powers of the signal coefficient.

6.4 Why Fisher information is supporting theory, not the headline

The statement that noising destroys fine or high-order information is general. The more distinctive contribution is algorithmic and structural:

Better headline

For a one-dimensional Markov distribution, the Bayes-optimal denoising score can be computed by exact message passing even when the marginal noisy density has no closed form; the unknown local law can then be learned by EM or energy-based maximum likelihood.

The information analysis should explain when the non-Gaussian gain is available, not replace the central result.

7 Neural denoisers and inductive bias

The benchmark included an MLP, windowed CNN, dilated CNN, bidirectional GRU, transformer, and one-dimensional U-Net. These architectures solve the same denoising task with different assumptions.

Model	Main inductive bias	Expected behaviour
MLP	Weak sequence structure	Must learn translation and locality from data
Window CNN	Locality and weight sharing	Strong in low-data regimes when correlations fit inside the window
Dilated CNN	Larger receptive field with few layers	Better for longer-range effects while remaining local
Bidirectional GRU	Sequential compression from both directions	Learned analogue of forward and backward information flow
Transformer	Global content-dependent interaction	Highly flexible but statistically expensive for a first-order process
1D U-Net	Multi-scale aggregation	Captures local and broad structure, but with many parameters

7.1 Why the window CNN can be strong

A small CNN has relatively few parameters and nearly the right locality bias. It can therefore beat larger models when data are scarce. This is a bias–variance trade-off:

- small window: more approximation bias, less estimation variance;
- large model: less approximation bias, more data and optimisation required.

7.2 Why privileged-input baselines should not be central

A model that receives the clean sequence as an input is not a deployable denoiser. Such an experiment may be retained as a diagnostic oracle, but should not be presented as a fair operational baseline.

8 Misspecification and hidden global structure

Suppose a global latent variable G affects the entire sequence:

$$G \in \{-1, +1\}, \qquad X_i = G + U_i,$$

where U_i is locally Markov. Conditioned on G, the process may be first-order Markov, but after marginalising G,

$$p(x_{i+1} \mid x_{1:i}) \neq p(x_{i+1} \mid x_i).$$

The entire history helps infer G.

A first-order model then converges, with infinite data, to the closest model inside its restricted family:

$$K^* = \arg\min_K D_{\mathrm{KL}}(p_{\mathrm{true}} \| p_K).$$

Estimation variance disappears, but approximation bias remains.

State augmentation

A non-Markov observed process can often be made Markov by enlarging the state. Here one could use $S_i = (X_i, G)$, leading naturally to a hidden Markov or latent-state extension.

This experiment is valuable because it separates failure due to insufficient data from failure due to an insufficient state representation.

9 Making BP faster and more flexible

9.1 Grid BP and its cost

If each message is stored at G grid points and the transition is dense, one update costs $O(G^2)$. A sequence of length L therefore costs

$$O(LG^2).$$

With $G = 401$, this can dominate inference.

9.2 Fourier acceleration

If

$$K(x' \mid x) = q(x' - x),$$

the predictive update is a convolution:

$$\widetilde{\alpha}_{i+1} = q * \alpha_i.$$

In Fourier space,

$$\widehat{\widetilde{\alpha}}_{i+1}(k) = \widehat{q}(k)\widehat{\alpha}_i(k).$$

For an autoregressive kernel $q(x' - \rho x)$, one obtains a scaled-frequency relation of the form

$$\widehat{\widetilde{\alpha}}_{i+1}(k) = \widehat{q}(k)\widehat{\alpha}_i(\rho k),$$

up to Fourier convention and interpolation details.

An FFT implementation can reduce convolutional updates to approximately

$$O(LG \log G).$$

A truncated rank-R representation may cost $O(LGR)$.

9.3 The likelihood multiplication problem

The transition step is diagonal in Fourier space, but the local observation update

$$\alpha_{i+1}(x') = g_\sigma(y_{i+1} \mid x')\widetilde{\alpha}_{i+1}(x')$$

is multiplication in real space and therefore convolution in Fourier space. A practical method may alternate between bases or use low-rank approximations.

9.4 Positivity and truncation

A BP message must be nonnegative. A truncated Fourier series can oscillate below zero. Possible remedies include:

- representing the log-message;
- positivity-preserving basis functions;
- spectral filtering;
- projection onto the nonnegative cone;
- low-rank factorizations of the transfer operator rather than direct message truncation.

Three different spectral ideas

Do not conflate Fourier modes used for computation, Hermite modes used to analyse Gaussian noising, and transfer-operator eigenfunctions used to describe stationary laws and correlation lengths.

9.5 Parametric non-Gaussian messages

A message could be approximated by a low-dimensional family $m(x;\eta_i)$ rather than a 401-point grid. Gaussian closure uses two parameters. A Gaussian-plus-Laplace ansatz might use a few more.

Exact closure requires that both

$$m(x) \mapsto m(x)g_\sigma(y \mid x)$$

and

$$m(x) \mapsto \int K(x' \mid x)m(x)\,\mathrm{d}x$$

remain inside the chosen family. Gaussian distributions have this property for linear Gaussian models. Laplace families generally do not: Gaussian convolution and Gaussian multiplication leave the Laplace family. Such a scheme should therefore be treated as a controlled approximation unless a special algebraic structure is proved.

9.6 A neural kernel inside BP

Instead of training a direct score network, train the local transition law:

$$K_\theta(x' \mid x) = \frac{\exp f_\theta(x,x')}{\int \exp f_\theta(x,u)\,\mathrm{d}u},$$

or use a mixture-density network. BP then composes the learned local rule globally.

This separates tasks cleanly:

- the neural network learns local physics;
- the probabilistic algorithm performs global inference;
- the same learned kernel can be reused across noise levels and missing-data patterns.

The main risks are narrow spikes, oscillatory kernels, grid under-resolution, and overfitting. Use-

ful regularisers include smoothness penalties, entropy control, symmetry constraints, and low-rank parameterisations.

9.7 Sampling from the learned model

If a normalised stationary kernel is known, sample ancestrally:

$$X_1 \sim \pi_\theta, \qquad X_{i+1} \sim K_\theta(\cdot \mid X_i).$$

For posterior sampling after a forward pass, sample $X_L \sim p(X_L \mid y)$ and then move backward with

$$p(x_i \mid x_{i+1}, y) \propto \alpha_i(x_i) K_\theta(x_{i+1} \mid x_i).$$

This is forward-filtering backward-sampling. Merely "starting from a flat marginal and running BP" is not, by itself, a complete sampling algorithm.

9.8 Runtime comparisons

A fair comparison should report:

- kernel-training time;
- neural-network training time;
- latency per sequence and per noise level;
- batch throughput;
- memory use;
- hardware, L, grid size G, and mixture size C;
- error-versus-runtime Pareto curves.

It is plausible that EM-BP is cheaper to train but slower per sample than an amortised CNN. The comparison must be measured rather than asserted.

Part II: One-Dimensional Gradient SPDEs as the Clean Benchmark

10 From a gradient SPDE to a Gibbs field

Consider

$$\partial_t \phi(t,x) = \partial_x^2 \phi(t,x) - V'(\phi(t,x)) + \sqrt{2\Theta}\,\eta(t,x),$$

where η is space–time white noise and Θ is physical temperature.

Define the energy functional

$$H[\phi] = \int \left[\frac{1}{2}(\partial_x \phi)^2 + V(\phi)\right] \mathrm{d}x.$$

Its variational derivative is

$$\frac{\delta H}{\delta \phi} = -\partial_x^2 \phi + V'(\phi).$$

Hence the SPDE is an infinite-dimensional overdamped Langevin equation:

$$\partial_t \phi = -\frac{\delta H}{\delta \phi} + \sqrt{2\Theta}\,\eta.$$

Under suitable boundary and regularity conditions, its reversible stationary measure is

$$\mu(\mathrm{d}\phi) \propto \exp\left[-\frac{H[\phi]}{\Theta}\right] \mathrm{d}\phi.$$

Interpretation

The deterministic drift lowers the energy, while the noise injects thermal fluctuations. Reversibility expresses detailed balance between these tendencies at equilibrium.

11 Spatial discretisation gives an exact nearest-neighbour model

Let $x_i = ih$ and $\phi_i = \phi(x_i)$. Approximate

$$\partial_x \phi(x_i) \approx \frac{\phi_{i+1} - \phi_i}{h}.$$

Then

$$\int \frac{1}{2}(\partial_x \phi)^2 \,\mathrm{d}x \approx \sum_i \frac{(\phi_{i+1} - \phi_i)^2}{2h},$$

and

$$\int V(\phi(x))\,\mathrm{d}x \approx \sum_i h V(\phi_i).$$

Therefore

$$H_h(\phi) = \sum_i \left[\frac{(\phi_{i+1} - \phi_i)^2}{2h} + hV(\phi_i)\right],$$

and

$$p(\phi_1, \dots, \phi_N) \propto \prod_i \exp\left[-\frac{(\phi_{i+1} - \phi_i)^2}{2\Theta h} - \frac{hV(\phi_i)}{\Theta}\right].$$

A symmetric pair energy is

$$W(\phi, \phi') = \frac{(\phi' - \phi)^2}{2\Theta h} + \frac{h}{2\Theta}[V(\phi) + V(\phi')],$$

so, for periodic boundaries,

$$p(\phi_{1:N}) \propto \prod_i e^{-W(\phi_i, \phi_{i+1})}.$$

The quadratic term acts like an elastic spring between neighbours; the local potential attracts each site toward preferred field values.

12 Two notions of spatial Markovity

12.1 Local Markov random field

Conditioned on its neighbours, an interior site is independent of the rest:

$$p(\phi_i \mid \phi_{-i}) = p(\phi_i \mid \phi_{i-1}, \phi_{i+1}).$$

Explicitly,

$$p(\phi_i \mid \phi_{i-1}, \phi_{i+1}) \propto \exp\left[-\frac{(\phi_i - \phi_{i-1})^2}{2\Theta h} - \frac{(\phi_{i+1} - \phi_i)^2}{2\Theta h} - \frac{hV(\phi_i)}{\Theta}\right].$$

12.2 Directed-chain representation

Because the graph is a chain, the same equilibrium law can be written as

$$p(\phi_{1:N}) = \pi(\phi_1) \prod_{i=1}^{N-1} P(\phi_{i+1} \mid \phi_i).$$

The transition P is not obtained by merely normalising $e^{-W(\phi,\phi')}$. It contains a future-continuation factor encoded by the leading transfer-operator eigenfunction.

13 Transfer matrices and belief propagation

Define the positive transfer kernel

$$K(\phi, \phi') = e^{-W(\phi,\phi')}.$$

As an integral operator,

$$(Kf)(\phi') = \int K(\phi, \phi') f(\phi)\, \mathrm{d}\phi.$$

On a field-value grid $z_1, \dots, z_G$, it becomes a matrix.

A forward message satisfies

$$\alpha_{i+1}(\phi') = \int \alpha_i(\phi) K(\phi, \phi')\, \mathrm{d}\phi,$$

with local observation factors inserted when denoising. A backward message obeys the reverse recursion. The one-site and pair marginals have exactly the same form as in the Markov-chain derivation.

Three names for one computation

On a one-dimensional chain, belief propagation, the forward–backward algorithm, and the transfer-matrix method are the same algebra expressed in probabilistic, signal-processing, and statistical-physics languages.

14 The ground-state transform

Suppose the symmetric positive operator K has leading eigenvalue λ_0 and positive eigenfunction ψ_0:

$$\int K(\phi, \phi')\psi_0(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi).$$

Define

$$P(\phi' \mid \phi) = \frac{K(\phi, \phi')\psi_0(\phi')}{\lambda_0\psi_0(\phi)}.$$

This is normalised because the numerator integrates to $\lambda_0\psi_0(\phi)$. If K is symmetric, the stationary density is

$$\pi(\phi) \propto \psi_0(\phi)^2.$$

Moreover,

$$\pi(\phi)P(\phi' \mid \phi) = \frac{1}{\lambda_0}\psi_0(\phi)K(\phi, \phi')\psi_0(\phi')$$

is symmetric, proving detailed balance.

For the nonsymmetric one-sided factorisation, the transition has the schematic form

$$P(\phi' \mid \phi) \propto \exp\left[-\frac{(\phi' - \phi)^2}{2\Theta h} - \frac{hV(\phi')}{\Theta}\right]\frac{\psi_0(\phi')}{\psi_0(\phi)}.$$

Meaning of the ground-state factor

The Gaussian term describes a local spatial increment. The potential rewards low-energy destinations. The ratio $\psi_0(\phi')/\psi_0(\phi)$ accounts for how many favourable future continuations are available after arriving at ϕ'.

For a finite open chain, the exact transition is position dependent and uses the backward message:

$$P_i(\phi' \mid \phi) = \frac{K(\phi, \phi')\beta_{i+1}(\phi')}{\beta_i(\phi)}.$$

Far from boundaries, β_i is dominated by the leading eigenfunction and the homogeneous ground-state transform emerges.

15 Why a fixed innovation model is misspecified

A simple innovation model has

$$\phi_{i+1} = \rho\phi_i + \varepsilon_i, \qquad P(\phi' \mid \phi) = q(\phi' - \rho\phi).$$

The conditional shape is fixed and only translated with ϕ.

The SPDE transition instead depends on the absolute destination through

$$e^{-hV(\phi')/\Theta}\psi_0(\phi').$$

For a double-well potential:

- near $+a$, the field prefers to remain in the positive well;
- near $-a$, it prefers the negative well;
- near zero, the continuation can branch toward either phase;
- near a kink, surrounding evidence makes the conditional asymmetric.

Therefore the conditional law changes shape with the current state. A fixed mixture of state-independent innovations is structurally misspecified. This motivates a learned state-dependent kernel or pair energy.

16 Learning a pair energy

Instead of normalising a conditional kernel separately for every ϕ, define

$$p_\theta(\phi_{1:N}) = \frac{1}{Z_\theta} \exp\left[-\sum_i W_\theta(\phi_i, \phi_{i+1})\right].$$

Benefits include:

- symmetry $W_\theta(\phi, \phi') = W_\theta(\phi', \phi)$ can enforce reversibility;
- global normalisation is tractable in one dimension;
- the same transfer operator yields Z_θ, marginals, samples, and gradients;
- the model aligns naturally with the underlying physics.

For a periodic grid model,

$$Z_\theta = \mathrm{tr}(K_\theta^N), \qquad K_\theta(\phi, \phi') = e^{-W_\theta(\phi,\phi')}.$$

For open boundaries, boundary vectors replace the trace.

16.1 Clean-data maximum likelihood

Let

$$E_\theta(\phi) = \sum_i W_\theta(\phi_i, \phi_{i+1}).$$

The negative log-likelihood gradient is

$$\nabla_\theta \mathcal{L} = \mathbb{E}_{\text{data}}[\nabla_\theta E_\theta] - \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta].$$

The first term is the positive phase and the second is the negative phase. Both reduce to adjacent-pair expectations.

16.2 Noisy-data learning and Fisher identity

For noisy observations, the first term becomes a posterior expectation:

$$\nabla_\theta \mathcal{L}_{\text{noisy}} = \mathbb{E}_{p_\theta(\phi|y)}[\nabla_\theta E_\theta(\phi)] - \mathbb{E}_{p_\theta(\phi)}[\nabla_\theta E_\theta(\phi)].$$

The posterior and equilibrium pair marginals are both available from transfer-matrix passes. This is exact one-dimensional Boltzmann learning, up to field-grid quadrature.

Gradient sign

The expression above is the gradient of the negative log-likelihood. The log-likelihood gradient has the opposite sign.

17 Exact denoising score for the field

17.1 Clean Gibbs score

For an interior coordinate,

$$s_i(\phi) = \frac{\partial}{\partial \phi_i} \log p(\phi).$$

Using the discretised energy,

$$s_i(\phi) = -\frac{1}{\Theta}\left[\frac{2\phi_i - \phi_{i-1} - \phi_{i+1}}{h} + hV'(\phi_i)\right].$$

The clean score combines a discrete Laplacian with a nonlinear local force.

17.2 Noisy score

With

$$Y_i = \phi_i + \sigma Z_i,$$

Tweedie's identity gives

$$\nabla_y \log p_\sigma(y) = \frac{\mathbb{E}[\phi \mid Y = y] - y}{\sigma^2}.$$

BP computes the posterior means, so the oracle noisy score is available across the entire noise schedule.

This gives a benchmark in which the exact target is not another neural estimator but a controlled transfer-matrix calculation.

18 The ϕ^4 double well, domains, and kinks

Take

$$V(\phi) = \frac{\lambda}{4}(\phi^2 - a^2)^2.$$

The minima are $\phi = \pm a$. The local potential encourages each site to choose a well, while the gradient term penalises rapid spatial changes. Low-temperature samples consist of long positive or negative domains separated by kinks.

A continuum kink is

$$\phi_{\text{kink}}(x) = a \tanh\left[a\sqrt{\frac{\lambda}{2}}(x - x_0)\right].$$

Its energy is

$$E_{\text{kink}} = \int_{-a}^{a} \sqrt{2V(u)}\,\mathrm{d}u = \frac{2\sqrt{2\lambda}}{3}a^3.$$

At low temperature, the kink density behaves approximately as

$$\rho_{\text{kink}} \asymp e^{-E_{\text{kink}}/\Theta},$$

up to prefactors, and the typical domain length scales as

$$\xi \asymp e^{E_{\text{kink}}/\Theta}.$$

18.1 Why Gaussian closure fails

A one-site marginal may be approximately

$$p(\phi_i) \approx \frac{1}{2}\mathcal{N}(+a, \tau^2) + \frac{1}{2}\mathcal{N}(-a, \tau^2).$$

Its mean is close to zero, even though zero is near the top of the potential barrier and is not a typical bulk value.

More importantly, the posterior can express two coherent hypotheses:

$$\text{the whole region is positive} \qquad \text{or} \qquad \text{the whole region is negative.}$$

A Gaussian approximation averages these into an unphysical field near zero. It may also smear a sharp kink over a broad region when the true uncertainty concerns kink position rather than kink shape.

19 Speciation as commitment to spatial structure

In diffusion models, speciation can be interpreted as the point at which a noisy trajectory or posterior commits to one among several structural possibilities.

For the ϕ^4 field, those possibilities are spatially coherent domains rather than independent sitewise signs. At high corruption noise σ:

- the posterior may remain ambiguous between positive and negative phases;
- kink locations are uncertain;

- the posterior mean can lie near zero.

As σ decreases:

- whole regions commit to $+a$ or $-a$;
- evidence propagates across correlated domains;
- kink positions localise;
- the denoiser changes from a mixture average to a structured configuration.

Two controls must remain distinct:

$$\Theta = \text{physical temperature of the target law}, \qquad \sigma = \text{artificial corruption level}.$$

Sweeping Θ changes the target's correlation length and multimodality. Sweeping σ traces the denoising problem for a fixed target.

20 No true one-dimensional phase transition, but a sharp crossover

For finite-range one-dimensional systems at every fixed $\Theta > 0$, domain walls have nonzero density. Hence there is no true finite-temperature spontaneous-symmetry-breaking transition of the higher-dimensional Ising type.

Nevertheless, the correlation length can be enormous:

$$\xi(\Theta) \sim e^{E_{\text{kink}}/\Theta}.$$

For a finite system of length L:

- if $L \ll \xi$, a sample often appears globally positive or globally negative;
- if $L \gg \xi$, it contains many domains.

Thus the model provides a controlled finite-size crossover from local fluctuations to apparently global commitment, without falsely claiming a thermodynamic phase transition.

21 Discretisation caveats

21.1 Each fine step is nearly Gaussian

The potential enters a one-step factor as

$$e^{-hV(\phi')/\Theta} = 1 - \frac{hV(\phi')}{\Theta} + O(h^2).$$

The typical increment is

$$\phi_{i+1} - \phi_i = O(\sqrt{h}).$$

Therefore a single very fine spatial step is nearly Gaussian. Nonlinearity accumulates across roughly ξ_{physical}/h sites.

This means one-step fit is not enough. The evaluation should include multi-step correlations, bimodality, kink statistics, domain lengths, and global score error.

A sensible discretisation resolves the kink width while retaining measurable nonlinear structure per transition. A robust procedure is to generate at fine resolution and subsample systematically.

21.2 Correlation length in sites

If ξ_{physical} is fixed,

$$\xi_{\text{sites}} = \frac{\xi_{\text{physical}}}{h}.$$

As $h \downarrow 0$, the same physical domain occupies more lattice sites.

A CNN with radius r sites has physical receptive field

$$R_{\text{physical}} = rh.$$

Keeping r fixed while refining the grid makes the physical receptive field shrink. To preserve physical coverage,

$$r \propto \frac{1}{h}.$$

Sharp experimental prediction

Under grid refinement, a fixed-window CNN should degrade relative to BP unless its receptive field is increased proportionally to $1/h$. BP's statistical range is not capped by a fixed local window.

22 Direct equilibrium sampling by the transfer matrix

Temporal SPDE simulation can equilibrate slowly at low temperature. The transfer matrix avoids burn-in.

Choose field values $z_1, \dots, z_G$ and quadrature weights $w_1, \dots, w_G$. Form the symmetric weighted matrix

$$\widetilde{K}_{ab} = \sqrt{w_a}\, e^{-W(z_a, z_b)} \sqrt{w_b}.$$

Compute its dominant eigenpair

$$\widetilde{K} v_0 = \lambda_0 v_0.$$

Then

$$\pi_a \propto v_{0,a}^2, \qquad P_{ab} = \frac{\widetilde{K}_{ab} v_{0,b}}{\lambda_0 v_{0,a}}.$$

Sample $\phi_1 \sim \pi$ and then sample sequentially from P.

"Exact" here means exact for the discretised and truncated field-value representation. Convergence should be checked by increasing the field range and refining the quadrature grid.

23 Why Fourier BP is especially natural for the SPDE

The symmetric transfer kernel factorises as

$$K(\phi, \phi') = D(\phi)G(\phi' - \phi)D(\phi'),$$

where

$$D(\phi) = e^{-hV(\phi)/(2\Theta)}, \qquad G(\phi' - \phi) = e^{-(\phi'-\phi)^2/(2\Theta h)}.$$

A message update becomes

$$m_{\text{new}}(\phi') = D(\phi') \int G(\phi' - \phi)D(\phi)m(\phi)\,\mathrm{d}\phi.$$

Thus one update is:

1. multiply by D in real space;
2. convolve with a Gaussian;
3. multiply by D again.

The convolution is diagonal in Fourier space. This makes the SPDE benchmark a particularly clean environment for testing the Fourier-BP idea discussed in the meeting.

24 The four-arm experiment

Use the double-well ϕ^4 potential on a fixed physical interval. Sweep physical temperature Θ, spatial spacing h, and denoising noise σ.

24.1 Arm 1: learned pair energy

Learn $W_\theta(\phi, \phi')$ from clean or noisy fields. Use BP for exact global inference. This tests whether a flexible local model can recover the nonlinear physics.

24.2 Arm 2: Gaussian closure

Retain only means and covariances. This should be competitive at high temperature or high noise, but fail when coherent phase ambiguity and kink uncertainty dominate.

24.3 Arm 3: window CNN

Train a local denoiser or score estimator. Its performance should depend strongly on the ratio between window size and correlation length in sites.

24.4 Arm 4: exact oracle score

Use the known pair energy and transfer-matrix posterior to calculate

$$s_\sigma(y) = \frac{\mathbb{E}[\phi \mid y] - y}{\sigma^2}.$$

This isolates error due to model misspecification, finite data, optimisation, numerical resolution, and restricted receptive fields.

24.5 Metrics

Report at least:

$$\mathbb{E}\|\widehat{s}_\sigma(Y) - s_\sigma(Y)\|^2, \qquad \mathbb{E}\|\widehat{\phi}(Y) - \phi\|^2,$$

and physical statistics:

$$C(r) = \mathbb{E}[\phi_i \phi_{i+r}],$$

correlation length, kink density, domain-length distribution, marginal bimodality, generated-sample quality, and runtime.

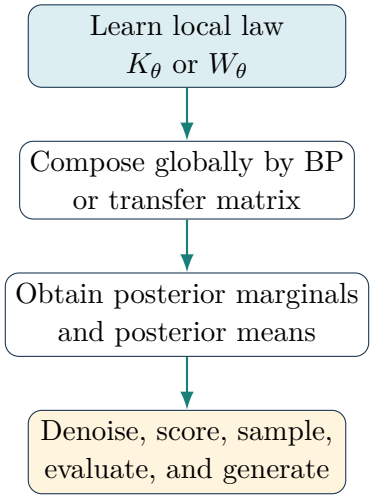
24.6 Expected qualitative pattern

Regime	Expected behaviour
High Θ	Short correlations and weak bimodality; all methods closer together
Intermediate Θ	Domains emerge; Gaussian closure starts losing structured information
Low Θ	Coherent phase and kink uncertainty dominate; structured BP gains the most
Smaller h	Fixed-window CNN deteriorates unless its window grows like $1/h$
More expressive W_θ	Learned model approaches the exact oracle if data and quadrature are sufficient

Part III: Unified Research Narrative

25 The unified interpretation

The original synthetic Markov experiment and the SPDE proposal are instances of the same decomposition:



A direct score network attempts to learn all of these operations jointly. The structured approach learns only the local law and delegates global composition to an exact algorithm.

Learn local physics, infer globally

This middle path lies between a fully known probabilistic model and a fully model-free neural denoiser. It can be statistically efficient, interpretable, reusable across noise levels, and exactly benchmarked in one dimension.

26 What can be claimed, and what should not be claimed

26.1 Defensible claims

- Exact posterior denoising is possible for continuous-state one-dimensional Markov models through BP, up to numerical quadrature.
- The Gaussian limit provides an analytic and algorithmic sanity check through the linear score and Kalman smoothing.
- EM or energy-based likelihood learning can estimate an unknown local law while preserving exact global inference.
- Non-Gaussian structure matters most where Gaussian smoothing has not yet erased it.
- Explicit structure can improve sample efficiency and robustness when the assumed state is correct.
- The ϕ^4 field provides a physically meaningful benchmark with exact transfer-matrix references, multimodality, kinks, and tunable correlation length.

26.2 Claims to avoid or qualify

- Do not claim universal superiority over neural networks without runtime, data-scale, and architecture controls.
- Do not call grid BP mathematically exact without stating the quadrature and truncation approximations.
- Do not call the full Gaussian forward–backward pass merely a Kalman filter; it is smoothing.
- Do not call the one-dimensional low-temperature crossover a true finite-temperature phase transition.
- Do not treat a Laplace message ansatz as exact without a closure proof.
- Do not equate a fixed innovation family with the general reversible SPDE transition.

27 A coherent paper structure

A compact paper could follow the logic below.

1. **Problem and structural thesis.** Explain why score estimation for Markov data reduces to posterior means.
2. **Exact inference.** Derive BP and Tweedie's identity.
3. **Learning.** Present EM for a transition kernel or Boltzmann learning for a pair energy.
4. **Sanity check.** Prove equivalence to Gaussian score and Kalman smoothing.
5. **Controlled non-Gaussian experiment.** Separate covariance and fourth-order information.
6. **Physical benchmark.** Introduce one-dimensional ϕ^4 and exact transfer-matrix references.
7. **Comparisons.** Evaluate Gaussian closure, local CNNs, larger networks, and the oracle.
8. **Computation.** Report grid, Fourier, and runtime trade-offs.
9. **Limitations.** Show global-latent misspecification and discretisation effects.

28 Figures-first plan

The meeting advice was to identify five or six decisive figures before expanding the prose.

No.	Figure	Scientific purpose
1	Method overview	Show local-law learning, BP posterior inference, Tweedie score, and optional EM loop
2	Gaussian sanity check	Demonstrate convergence of grid BP to analytic score and Kalman smoother
3	Non-Gaussian gain	Compare exact BP, EM-BP, Gaussian score, and neural denoisers across noise levels
4	Sample efficiency	Plot error versus number of training sequences; reveal inductive-bias advantage
5	SPDE temperature sweep	Show domains, kink density, correlation length, and score error across Θ
6	Refinement and runtime	Compare fixed-window CNN and BP across h, with runtime–accuracy Pareto curves
7	Misspecification appendix	Show irreducible bias under a global latent and repair by state augmentation

29 Practical next steps

1. Verify the current numerical results and recover exact metric definitions from frozen outputs.
2. Harmonise notation throughout the code, report, paper, and thesis.
3. Add the Gaussian score / Kalman-smoother proof as a short proposition.
4. Profile dense grid BP and produce baseline wall-clock measurements.
5. Implement FFT-based Gaussian convolution for the current Markov model.
6. Build the ϕ^4 transfer operator and verify quadrature convergence.
7. Generate exact equilibrium samples by the leading-eigenvector transform.
8. Compute oracle posterior scores for a small (Θ, h, σ) grid.
9. Reuse the existing Gaussian closure and CNN code as the first comparison arms.
10. Only after the exact benchmark is stable, introduce a learned pair energy or neural kernel.

Minimal decisive experiment

Start with known V, exact transfer-matrix sampling, and exact score. Compare only the Gaussian closure and the existing window CNN across temperature and grid spacing. This already tests multimodality, domain-scale commitment, and receptive-field scaling before adding the complexity of learning W_θ.

A Detailed derivation of the Gaussian score identity

Let

$$A = \Sigma_X + \sigma^2 I.$$

The posterior mean is

$$\mathbb{E}[X \mid y] = \Sigma_X A^{-1} y.$$

Then

$$\begin{aligned} \mathbb{E}[X \mid y] - y &= \Sigma_X A^{-1} y - A A^{-1} y \\ &= (\Sigma_X - A) A^{-1} y \\ &= -\sigma^2 A^{-1} y. \end{aligned}$$

Dividing by σ^2 yields

$$\frac{\mathbb{E}[X \mid y] - y}{\sigma^2} = -A^{-1} y = \nabla_y \log \mathcal{N}(y; 0, A).$$

This identity is exact in any dimension.

B Forward–backward pseudocode

Grid BP for one noisy sequence

Inputs: observations $y_{1:L}$, state grid $z_{1:G}$, quadrature weights $w_{1:G}$, transition matrix K, likelihood values $g_i(a) = g_\sigma(y_i \mid z_a)$.

Forward pass

1. $\alpha_1(a) \leftarrow \mu(a) g_1(a)$; normalise.
2. For $i = 1, \dots, L-1$:

$$\widetilde{\alpha}_{i+1}(b) \leftarrow \sum_a \alpha_i(a) K_{ab} w_a,$$

$$\alpha_{i+1}(b) \leftarrow g_{i+1}(b) \widetilde{\alpha}_{i+1}(b);$$

   normalise and store the scaling constant.

Backward pass

1. $\beta_L(a) \leftarrow 1$.
2. For $i = L-1, \dots, 1$:

$$\beta_i(a) \leftarrow \sum_b K_{ab} g_{i+1}(b) \beta_{i+1}(b) w_b;$$

   normalise consistently.

Posterior

$$\gamma_i(a) \propto \alpha_i(a) \beta_i(a), \qquad \widehat{x}_i = \sum_a z_a \gamma_i(a) w_a.$$

Score

$$\widehat{s}_i(y) = \frac{\widehat{x}_i - y_i}{\sigma^2}.$$

C EM pseudocode

EM with latent clean states

Initialise $\theta^{(0)}$.
For iterations $r = 0, 1, 2, \ldots$:

1. **E-step.** For every noisy training sequence, run BP under $K_{\theta^{(r)}}$ and compute pair marginals
$$\xi_i^{(m)}(x, x') = p_{\theta^{(r)}}(X_i = x, X_{i+1} = x' \mid Y^{(m)}).$$
2. **M-step.** Maximise
$$Q(\theta \mid \theta^{(r)}) = \sum_{m,i} \mathbb{E}_{\xi_i^{(m)}}[\log K_\theta(X_{i+1} \mid X_i)].$$
3. Stop when the observed-data likelihood or held-out noisy objective stabilises.

D Pair-energy learning pseudocode

Exact one-dimensional Boltzmann learning

For each optimisation step:

1. Run posterior BP for every noisy data sequence and accumulate
$$G_{\text{data}} = \mathbb{E}_{p_\theta(\phi|y)}[\nabla_\theta E_\theta(\phi)].$$
2. Run an equilibrium transfer-matrix pass and accumulate
$$G_{\text{model}} = \mathbb{E}_{p_\theta(\phi)}[\nabla_\theta E_\theta(\phi)].$$
3. Update for negative log-likelihood minimisation:
$$\theta \leftarrow \theta - \eta(G_{\text{data}} - G_{\text{model}}).$$
4. Monitor likelihood, pair-statistic fit, quadrature convergence, and kernel smoothness.

E Notation glossary

Symbol	Meaning
X_i	Clean Markov-chain state
Y_i	Noisy observation
σ	Gaussian corruption scale
$K_\theta(x' \mid x)$	Normalised transition kernel
$W_\theta(x, x')$	Pair energy
α_i, β_i	Forward and backward messages
γ_i	One-site posterior marginal
ξ_i	Adjacent pair posterior marginal

$s_\sigma(y)$	Noisy-data score $\nabla_y \log p_\sigma(y)$
$u_\theta(y)$	Parameter score $\partial_\theta \log p_\theta(y)$
ϕ_i	Spatial field value at lattice site i
h	Spatial grid spacing
Θ	Physical temperature of the SPDE equilibrium law
$V(\phi)$	Local potential
$H[\phi]$	Continuum energy functional
ψ_0, λ_0	Leading transfer-operator eigenfunction and eigenvalue
ξ	Physical or lattice correlation length, according to context
E_{kink}	Domain-wall energy

F Meeting-derived action list

The discussion produced the following practical priorities:

1. Select five or six decisive figures and construct a summary figure for the paper opening.
2. Try truncated Fourier or low-rank representations for BP messages.
3. Relabel information plots by second- and fourth-order statistical modes rather than only by model parameters.
4. Add a concise proof of the Gaussian-score and Kalman equivalence.
5. Harmonise transition, message, score, and noise notation.
6. Recover and report actual inference-time comparisons.
7. Avoid making causal attention on Markov data the main novelty.
8. Explore a flexible neural kernel or pair energy inside exact BP.
9. Treat Laplace or mixed parametric messages as an open approximation question unless closure is proved.
10. Bring all collaborators into the scientific framing before locking the paper narrative.

G Final mental model

The whole story

A clean one-dimensional system is specified by a local transition or pair energy. Gaussian corruption creates a latent-chain posterior. BP computes exact posterior means and therefore exact denoising scores. EM or Boltzmann learning estimates the local law when it is unknown. Gaussian closure provides a useful but limited approximation; neural networks provide flexible amortisation but must learn the relevant structure from data. One-dimensional gradient SPDEs supply a physically meaningful benchmark where all of these components can be evaluated against an exact transfer-matrix reference.

End of document